Computer Networking Fundamentals

Study Notes: OSI Model, TCP/IP, IP Addressing, and Routing

1. The OSI Model

The Open Systems Interconnection (OSI) model is a conceptual framework that standardizes the functions of a telecommunication or computing system into seven distinct layers. It was developed by the International Organization for Standardization (ISO) to promote interoperability between different vendors' systems.

Layer	Name	Function	Example Protocols
7	Application	Provides network services directly to end-user applications	HTTP, FTP, SMTP, DNS
6	Presentation	Translates, encrypts, and compresses data	SSL/TLS, JPEG, ASCII
5	Session	Establishes, manages, and terminates sessions	NetBIOS, RPC
4	Transport	Ensures reliable data transfer, flow control	TCP, UDP
3	Network	Handles logical addressing and routing	IP, ICMP, OSPF
2	Data Link	Handles physical addressing (MAC) and error detection	Ethernet, PPP, ARP
1	Physical	Transmits raw bit stream over physical medium	Cables, Hubs, NICs

A common mnemonic for remembering the layers from top to bottom (7 to 1) is "All People Seem To Need Data Processing."

2. The TCP/IP Model

The TCP/IP model is a more practical, four-layer model used in real-world networking, including the internet. It combines several OSI layers into broader categories.

- **Application Layer** — combines OSI's Application, Presentation, and Session layers
- **Transport Layer** — equivalent to OSI Layer 4 (TCP and UDP)
- **Internet Layer** — equivalent to OSI Layer 3 (IP addressing and routing)
- **Network Access Layer** — combines OSI's Data Link and Physical layers

2.1 TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented (three-way handshake)	Connectionless
Reliability	Reliable, guarantees delivery and order	Unreliable, no delivery guarantee

Speed	Slower due to overhead	Faster, minimal overhead
Use Cases	Web browsing, email, file transfer	Video streaming, VoIP, DNS queries
Header Size	20 bytes minimum	8 bytes

The TCP three-way handshake consists of three steps: SYN, SYN-ACK, and ACK. This process establishes a reliable connection before any application data is transmitted.

3. IP Addressing

An IP address is a unique numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication.

3.1 IPv4

IPv4 addresses are 32 bits long, written as four decimal numbers (octets) separated by dots, each ranging from 0 to 255 (e.g., 192.168.1.1). This provides approximately 4.3 billion unique addresses, which has led to address exhaustion concerns.

Class	Range (First Octet)	Default Subnet Mask	Typical Use
A	1 - 126	255.0.0.0 (/8)	Large networks
B	128 - 191	255.255.0.0 (/16)	Medium-sized networks
C	192 - 223	255.255.255.0 (/24)	Small networks
D	224 - 239	N/A	Multicast
E	240 - 255	N/A	Experimental/Reserved

3.2 Private IP Address Ranges

Certain IP ranges are reserved for private networks (defined in RFC 1918) and are not routable on the public internet:

- Class A: 10.0.0.0 to 10.255.255.255
- Class B: 172.16.0.0 to 172.31.255.255
- Class C: 192.168.0.0 to 192.168.255.255

3.3 IPv6

IPv6 addresses are 128 bits long, written as eight groups of four hexadecimal digits separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334). IPv6 was designed to solve the address exhaustion problem of IPv4, offering approximately 340 undecillion unique addresses.

3.4 Subnetting

Subnetting divides a large network into smaller sub-networks (subnets) to improve performance and security. The subnet mask determines which portion of an IP address refers to the network and which refers to the host. A /24 subnet mask (255.255.255.0) allows for 254 usable host addresses, since one address is reserved for the network identifier and one for the broadcast address.

4. Networking Devices

Device	OSI Layer	Function
Hub	Physical (1)	Broadcasts data to all connected ports; no traffic filtering
Switch	Data Link (2)	Forwards data based on MAC address; creates separate collision domains
Router	Network (3)	Forwards packets between different networks based on IP address
Firewall	Network/Transport (3-4)	Filters traffic based on security rules
Access Point	Data Link (2)	Provides wireless connectivity to a wired network

5. DNS (Domain Name System)

DNS translates human-readable domain names (like www.example.com) into machine-readable IP addresses. It functions as a hierarchical, distributed database.

- **A Record** — maps a domain name to an IPv4 address
- **AAAA Record** — maps a domain name to an IPv6 address
- **CNAME Record** — maps an alias to another domain name
- **MX Record** — specifies the mail server for a domain
- **NS Record** — specifies the authoritative name servers for a domain

DNS operates primarily over UDP port 53 for standard queries, though it falls back to TCP port 53 for larger responses such as zone transfers.

6. Common Well-Known Ports

Port	Protocol	Service
20/21	TCP	FTP (Data/Control)
22	TCP	SSH
23	TCP	Telnet
25	TCP	SMTP
53	TCP/UDP	DNS
80	TCP	HTTP
110	TCP	POP3
143	TCP	IMAP
443	TCP	HTTPS

3389	TCP	RDP

7. Routing Basics

Routing is the process of selecting a path for traffic across networks. Routers use routing tables and routing protocols to determine the best path for forwarding packets.

- **Static Routing** — manually configured routes; not adaptive to network changes
- **Dynamic Routing** — routes are learned automatically via protocols like OSPF, RIP, and BGP
- **Default Route** — the route used when no other matching route is found in the routing table

Border Gateway Protocol (BGP) is the routing protocol used to exchange routing information between different autonomous systems on the internet, making it the backbone protocol of internet routing.